## RQ4 GPU coverage: which queries run on each GPU engine, by scale

### ◆ Polars GPU

| | SF 1 | SF 10 | SF 30 | SF 100 | SF 300 |
|---|---|---|---|---|---|
| **Baseline suite** | | | | | |
| q01 | ✓ | ✓ | ✓ | ✓ | ✓ |
| q03 | ✓ | ✓ | ✓ | ✓ | ✓ |
| q06 | ✓ | ✓ | ✓ | ✓ | ✓ |
| q10 | ✓ | ✓ | ✓ | ✓ | ✓ |
| q12 | ✓ | ✓ | ✓ | ✓ | ✓ |
| q14 | ✓ | ✓ | ✓ | ✓ | ✓ |
| **Complex suite** | | | | | |
| cx_q09 | – | – | ✓ | ✓ | ✓ |
| cx_q13 | – | – | ✗ | ✗ | ✗ |
| cx_semi_anti | – | – | ✓ | ✓ | ✓ |
| cx_window | – | – | ✗ | ✗ | ✗ |
| cx_distinct | – | – | ✓ | ✓ | ✓ |
| cx_self_join | – | – | ✓ | ✓ | ✓ |

### ▲ Spark GPU

| | SF 1 | SF 10 | SF 30 | SF 100 | SF 300 |
|---|---|---|---|---|---|
| **Baseline suite** | | | | | |
| q01 | ✓ | ✓ | ✓ | ✓ | ✓ |
| q03 | ✓ | ✓ | ✓ | ✓ | ✓ |
| q06 | ✓ | ✓ | ✓ | ✓ | ✓ |
| q10 | ✓ | ✓ | ✓ | ✓ | ✓ |
| q12 | ✓ | ✓ | ✓ | ✓ | ✓ |
| q14 | ✓ | ✓ | ✓ | ✓ | ✓ |
| **Complex suite** | | | | | |
| cx_q09 | – | – | ✓ | ✓ | ✓ |
| cx_q13 | – | – | ✓ | ✓ | ✓ |
| cx_semi_anti | – | – | ✓ | ✓ | ✓ |
| cx_window | – | – | ✓ | ✓ | ✓ |
| cx_distinct | – | – | ✓ | ✓ | ✓ |
| cx_self_join | – | – | ✓ | ✓ | ✓ |

| | | | |
|---|---|---|---|
| ✓ | Supported (all rounds) | ◐ k/n | Unstable (some rounds, k/n) |
| ✗ | Not supported on GPU | OOM | Out of memory |
| ! | Other error (integrity failure) | – | Not run at this scale |

Black outline: failure not pre-declared (design section 8 pre-declares cx_q13 and cx_window on Polars GPU). Not supported (✗) and out of memory (OOM) are coverage outcomes (A1); hatched: unstable coverage, a coverage outcome in some rounds only, excluded from every contrast in all rounds. Other error (!) is an integrity failure: it fails its round (section 7), so an accepted round should show none.